

HiScan™ SQ System

Two proven technologies, one powerful platform—arrays and sequencing in a single system.

HiScanSQ Highlights

- Designed for Flexibility**
 Get the high-throughput processing of arrays with the power and resolution of next-generation sequencing to enable a wide breadth of applications
- Proven Performance**
 Access the most successful, trusted array and sequencing technologies available
- One Streamlined Solution**
 Minimize instrument hands-on time through a number of cutting-edge user interface features that automate and streamline experiment setup and processing procedures

Introduction

The HiScanSQ system (Figure 1) integrates the high-throughput capacity of genotyping and gene expression arrays with the power and resolution of next-generation sequencing, delivering unprecedented flexibility for experimental design. The instrument features two distinct components: the HiScan Reader, a high-performance scanner, and the SQ Module, a fluidics device for genomic-scale sequencing. The HiScan Reader functions independently as a high-speed, precision microarray scanner for Illumina BeadArray™ products. With the SQ Module, the system can perform Illumina sequencing by synthesis (SBS) chemistry, the world's most widely adopted next-generation sequencing technology.

The integration of microarray and sequencing experiments offers unlimited possibilities for genetic analysis research. Together, these complementary technologies provide a powerful approach to genetic discovery and validation, enabling researchers to take any course of study and ask virtually any question of the genome, transcriptome, or epigenome. HiScanSQ allows researchers to:

- Identify target SNPs from sequencing experiments to design iSelect® custom genotyping panels for any species and any study
- Leverage targeted resequencing as the ultimate follow-on to genome-wide association studies (GWAS)
- Transition between array-based and sequencing-based gene expression analysis
- Validate array-based expression and splice variant data

Figure 1: The HiScanSQ System



The HiScanSQ system includes two separate components: the HiScan Reader (left) and the SQ Module (right). The HiScan Reader is a high-performance scanner, capable of analyzing Illumina BeadArray products and next-generation sequencing flow cells. The SQ module is an advanced fluidics device for SBS reagent handling.

BeadArray Technology

Illumina's BeadArray products provide powerful array-based solutions for genotyping, gene expression profiling, and methylation analyses. The proven GoldenGate® and Infinium® Genotyping assays offer the industry's best data quality and have catalyzed many revolutionary discoveries in disease research. The DASL® and DirectHyb Assays enable highly accurate genome-wide expression profiling from a range of sample types. HiScanSQ provides high-precision microarray scanning for the BeadChips associated with these assays. The instrument delivers high signal-to-noise ratios, low limits of detection, and a broad dynamic range to deliver superior results for a diverse number of applications.

High-Performance Array Scanner

The HiScan Reader is capable of sub-micron resolution using high-performance lasers, optics, and detection systems (Table 1). The result is dramatically reduced scan times, even on the highest density BeadChips (Table 2), while maintaining exceptionally high data quality and reproducibility.

Microarrays are making greater contributions in a growing number of fields, especially human disease research and trait studies for many agriculturally important species. As the technology continues to expand, the next generation of microarrays will feature increasingly higher numbers of markers per sample for whole-genome analysis. HiScanSQ* will be required to read these ultra-high density BeadChips and provide the high-quality data and rapid assay turnaround that Illumina customers expect.

* The iScan system can also be used for high-density BeadChip scanning

Table 1: HiScan Reader Performance Parameters

Performance Specification	
Lasers	A two-laser system with wavelengths at 532 and 660 nm
Optical System	Time Delayed Integration (TDI) line scanning and two CCD sensors for high-resolution performance and fast data rate
Scanner Resolution	0.375 micron spatial resolution with dual-channel collection

Table 2: HiScan Reader Array Scan Times

BeadChip	Scan Time (per sample)	Manual Loading (samples/week)	Automated Loading (samples/week)
HumanOmni1-Quad	12.5 minutes	160	576
HumanOmniExpress	5 minutes	480	1,440
Human660W-Quad	9 minutes	160	800
HumanCytoSNP-12	2.9 minutes	480	1,720
iSelect® HD	1.7 minutes	480	3,456
Universal-32*	0.3 minutes	1,440	1,440

* For GoldenGate Indexing™ assay

Sequencing by Synthesis

Illumina's next-generation sequencing technology has made genome-scale sequencing routine for labs of any size. This powerful research method enables hypothesis-free experimental design to observe cellular processes at an exquisite level of resolution, allowing scientists to push the frontier of biological discovery. Researchers have embraced its broad range of applications, and are publishing ground-breaking papers at an unprecedented rate (see www.illumina.com/publications).

Fast, Simple, Automated

As with all Illumina sequencing systems, HiScanSQ provides the industry's most streamlined workflow, allowing researchers to generate meaningful data quickly and efficiently (Figure 2 and Table 3). Libraries can be prepared from virtually any nucleic acid sample in less than six hours with standardized, ready-to-use kits. Clonal amplification requires less than four hours, with less than 10 minutes of hands-on time, using the fully automated cBot cluster generation system. This easy-to-use protocol does not require emulsion PCR. It provides a self-contained system that minimizes handling errors and contamination concerns, eliminating the need for robotics or clean rooms.

HiScanSQ includes a number of carefully designed features that offer a new level of simplicity and minimize the hands-on time for user interaction (Figure 3). A convenient touch screen provides an intuitive interface for rapid setup, including on-screen step-by-step instructions with embedded multimedia help. Pre-configured, plug-and-play

reagents can be simultaneously loaded into the instrument's chiller compartment, requiring only two minutes of hands-on time. Real-time progress indicators provide at-a-glance status reports, and remote monitoring allows progress checks on multiple systems from any browser or internet-enabled phone.

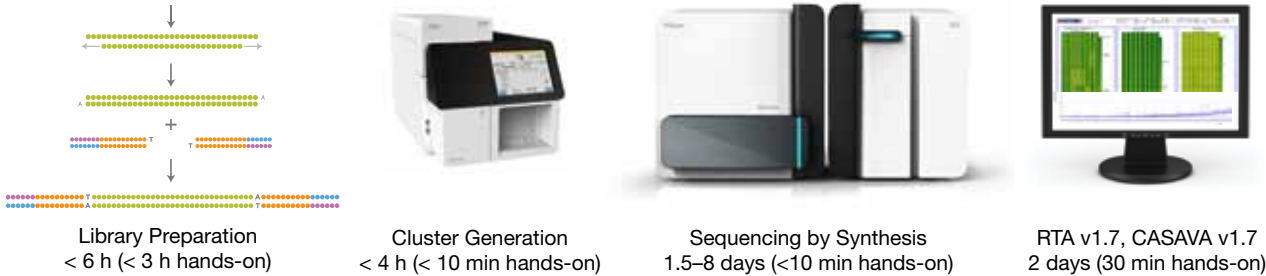
Add Sequencing Capabilities at Any Time

The SQ Module can be easily added to an existing HiScan Reader at any time to enable sequencing capabilities (Table 4). This compact device provides all the components necessary to perform Illumina's SBS chemistry, including an integrated paired-end fluidics system. This eliminates the need for a separate external module, and provides seamless second-read sequencing. Any lab interested in adding sequencing to their research can have greater control over experiments and scheduling without having to rely on large centralized instruments.

Diverse Applications

HiScanSQ enables a plethora of array- and sequencing-based applications (Table 5). The high-performance reader supports a number of proven array-based genotyping, gene expression profiling, and methylation assays. The combination of both short and long-insert paired-end sequencing, coupled with 2×100 bp read lengths and powerful SBS chemistry greatly expands the range of sequencing experiments. Low input requirements for both technologies enable a number of applications where sample is limited. With integrated sequencing and array technology, HiScanSQ opens up a world of new possibilities.

Figure 2: Next-Generation Sequencing Simplified



The industry's simplest sequencing workflow, from standardized library preparation kits, to automated cluster generation, to streamlined sequencing by synthesis and real-time data analysis.

Figure 3: HiScanSQ Innovative Design Features

The touch screen user interface facilitates step-by-step setup and provides real-time reporting



The combo drawer contains dual compartments for easy array and flow cell loading



Pre-configured, plug-and-play reagents are ready for up to 200 sequencing cycles



Internal paired-end fluidics allows for seamless second-read sequencing

Innovative engineering and human interaction design features provide a simplified user experience and minimize hands-on time.

Table 3: HiScanSQ Sequencing Application Examples

Read Length	Small Genome Sequencing	Targeted Resequencing	mRNA-Seq (Discovery)	mRNA-Seq (Profiling)	ChIP-Seq	Small RNA Sequencing
Example	4 Mb bacterium > 50x coverage	5 Mb region > 75x coverage	40M* reads per sample	Avg. 5M reads per sample	Avg. 2M reads per sample	2.5M reads per sample
Read Length	2 × 50 bp	2 × 75 bp	2 × 75 bp	1 × 50 bp	1 × 35 bp	1 × 35 bp
Samples per Run	96	96	6	48	96	96
Run Time	≤ 4.5 days	≤ 6.5 days	≤ 6.5 days	≤ 4.5 days	≤ 2.5 days	≤ 2.5 days

* Double the reads for paired-end runs

Table 4: HiScanSQ Sequencing Performance Parameters

	Performance Specification
Data Generation	> 50 Gb of high-quality data per paired-end flow cell
Run Times	1.5 days for a 36-bp single-read run 4 days for a 2 × 50-bp paired-end run 8 days for a 2 × 100-bp paired-end run
Daily Throughput	Minimum of 6 Gb of high-quality filtered bases per day
Samples per Flow Cell	1–12 samples per flow cell lane (8–96 samples per flow cell)
Cluster Density	260,000–347,000 clusters/mm ² 250M clusters passing filter per flow cell

Table 5: Diverse Array and Sequencing Applications

Research Areas	BeadArray Applications	Sequencing Applications
Whole-Genome Analysis	Whole-Genome Genotyping	Whole-Genome Discovery
Copy Number Variation (CNV)	CNV Analysis	CNV Discovery
Targeted Genome Analysis	Custom and Focused Genotyping	Targeted Resequencing
Gene Regulation and Epigenetic Analysis	Whole-Genome DNA Methylation Profiling	ChIP-Seq, Small RNA Analysis, Methylation Discovery and Analysis
Gene Expression	Whole-Genome Expression Profiling, FFPE Sample Analysis	Transcriptome Discovery, Profiling
Cytogenetics	Cytogenetic Analysis	Digital Karyotyping

